

UNIT 5 DATA WAREHOUSING

Data warehousing Components –Building a Data warehouse –Data Warehouse Architecture – DBMS Schemas for Decision Support – Data Extraction, Cleanup, and Transformation Tools –Metadata – reporting – Query tools and Applications – Online Analytical Processing (OLAP) – OLAP and Multidimensional Data Analysis.

Data Warehousing is the **process of collecting and storing large amounts** of data from different sources in one central place. This stored data is organized so that it can be easily analyzed to help businesses make better decisions.

Data warehousing is a critical component of **business intelligence and data analysis**. A data warehouse is a **centralized repository** where data from multiple sources is stored, transformed, and made available for analysis and reporting. Here are the main components of a data warehouse:

Characteristics of Data warehouse

- Subject-Oriented
- Integrated
- Time-variant
- Non-volatile

Subject-Oriented

A data warehouse is subject oriented as it offers information regarding a theme instead of companies' on-going operations. These subjects can be sales, marketing, distributions, etc. A data warehouse never focuses on the on-going operations.

Integrated

A data warehouse is developed by integrating data from varied sources like a mainframe, relational databases, flat files, etc.

Time Variant:

The data collected in a data warehouse is recognized with a particular period and offers information from the historical point of view. It contains an element of time, explicitly or implicitly. One such place where Data warehouse data display time variance is in the structure of the record key. Every primary key contained with the DW should have either implicitly or explicitly an element of time. Like the day, week month, etc. Another aspect of time variance is that once data is inserted in the warehouse, it can't be updated or changed.

Non-volatile

Data warehouse is also non-volatile means the previous data is not erased when new data is entered in it. Data is read-only and periodically refreshed. This also helps to analyze historical data and understand what & when happened. It does not require transaction process, recovery and concurrency control mechanisms. Activities like delete, update, and insert which are performed in an operational application environment are omitted in Data warehouse environment. Only two types of data operations performed in the Data Warehousing are

1. Data loading
2. Data access

KEY COMPONENTS of Data Warehousing

1. Data Sources

- **Internal Data Sources:** These are sources within an organization, such as transactional databases, operational databases, CRM, ERP, and other internal applications.

- **External Data Sources:** These include data from external sources, such as market research, social media, partner data, and third-party data providers.

2. ETL (Extract, Transform, Load) Processes

- **Extraction:** Data is collected from multiple sources and extracted into a staging area.
- **Transformation:** Data is cleaned, formatted, and transformed to ensure consistency, handle missing values, remove duplicates, and align with the data warehouse schema.
- **Loading:** The processed data is loaded into the data warehouse for storage.

3. Staging Area

The staging area is like a temporary storage space used in data warehousing. It is more like a "waiting room" where raw data from different sources is collected before it gets processed or transformed.

4. Data Storage

- **Data Warehouse:** The main repository for storing processed and structured data.
- **Data Marts:** Smaller, more focused repositories derived from the data warehouse, often built for **specific departments or business units**.
- **Metadata Repository:** Stores metadata, which includes information about the structure, definitions, and mapping of data. Metadata enables easier data management and helps users understand the data.

5. Data Access Tools

- **Query Tools:** Allow users to query and retrieve data from the data warehouse, typically through SQL-based tools.

- **Reporting and Analysis Tools:** Generate reports, dashboards, and visualizations, and provide insights for business intelligence.
- **OLAP (Online Analytical Processing):** Supports multidimensional analysis of data and enables complex queries that are necessary for decision-making.

6. Data Warehouse Management

- **Data Quality Management:** Ensures data accuracy, consistency, and reliability.
- **Data Security:** Manages access control, authentication, and data protection measures.
- **Performance Monitoring and Tuning:** Optimizes query performance, load balancing, and data retrieval processes for faster data access.

7. Data Warehouse Users

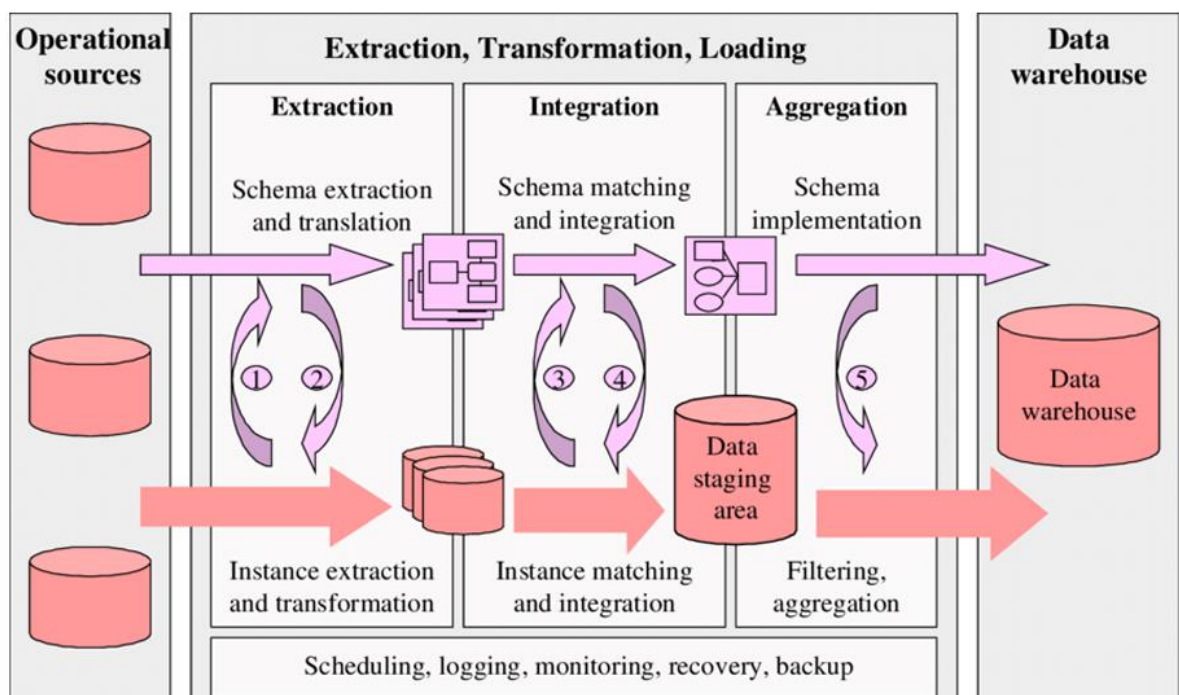
- **Business Analysts:** Use data for analytical purposes and reporting.
- **Data Scientists:** Perform in-depth analysis, statistical modeling, and predictive analytics.
- **Executives and Decision-Makers:** Rely on insights from the data warehouse for strategic decisions.
- **Data Engineers:** Manage ETL processes, maintain data quality, and ensure efficient data flow.

Each component works together to enable data collection, integration, and analysis for better data-driven decision-making across the organization.

BUILDING A DATA WAREHOUSE

In general, building any data warehouse consists of the following steps:

1. Extracting the transactional data from the data sources into a staging area.
2. Transforming the transactional data.
3. Loading the transformed data into a dimensional database.
4. Building pre-calculated summary values to speed up report generation.
5. Building (or purchasing) a front-end reporting tool.



Extracting Transactional Data:

A large part of building a DW is pulling data from various data sources and placing it in a central storage area.

Transforming Transactional Data:

An equally important and challenging step after extracting is transforming and relating the data extracted from multiple sources.

Creating a Dimensional Model:

The third step in building a data warehouse is coming up with a dimensional model. Most modern transactional systems are built using the relational model. The relational database is highly normalized; when designing such a system, you try to get rid of repeating columns and make all columns dependent on the primary key of each table. The relational systems perform well in the On-Line Transaction Processing (OLTP) environment.

Loading the Data:

Keep in mind that such data transformations can be performed at either of the two stages: while extracting the data from their origins or while loading data into the dimensional model.

Generating Pre-calculated Summary Values:

The next step is generating the precalculated summary values which are commonly referred to as aggregations(combinations). This step has been tremendously simplified by SQL Server Analysis Services (or OLAP Services, as it is referred to in SQL Server 7.0).

Building (or Purchasing) a Front-End Reporting Tool

The cost of building a custom reporting (and OLAP) tool will usually outweigh the purchase price of a third-party tool. That is not to say that OLAP tools are cheap. There are several major vendors on the market that have top-notch analytical tools. In addition to the third-party tools, Microsoft has just released its own tool, Data Analyzer, which can be a cost-effective alternative.

Data Warehouse Architecture

A Data Warehouse therefore can be described as a system that consolidates and manages data from different sources to assist an organization in making proper decisions. This makes the work of handling data to report easier. Two main construction approaches are used: Two of the most common models that have

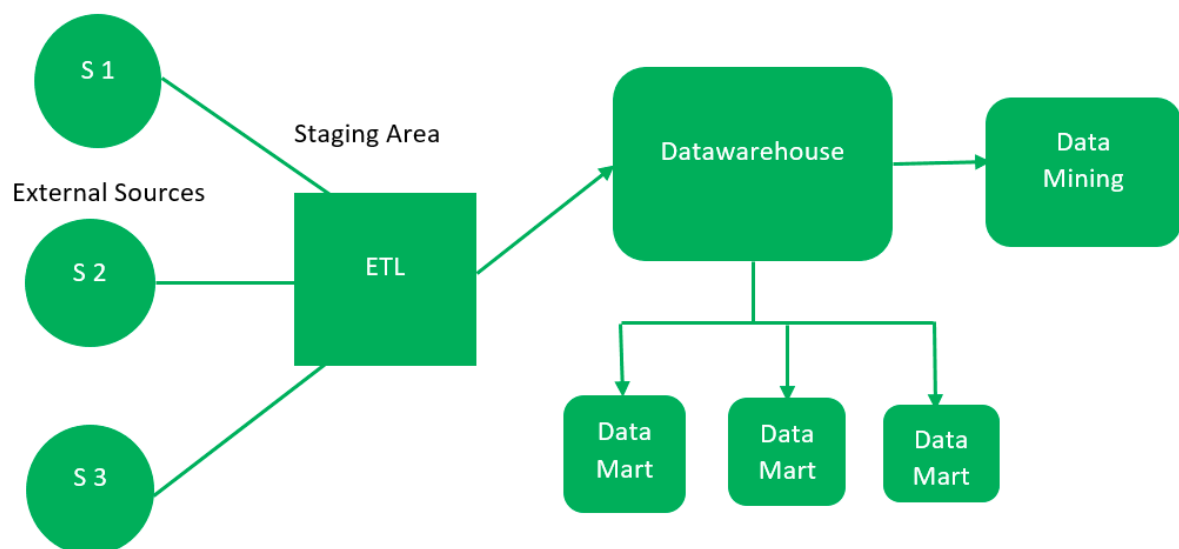
been developed are the Top-Down approach and the Bottom-Up approach and each of them possesses its strengths and weaknesses.

A **Data-Warehouse** is a heterogeneous collection of data sources organized under a unified schema. There are 2 approaches for constructing a data warehouse: The top-down approach and the Bottom-up approach

Top-Down Approach

The initial approach developed by Bill Inmon known as the top-down approach starts with building a single source data warehouse for the whole company.

Merges and processes external data through the ETL ([Extract, Transform, Load](#)) process and subsequently stores them in the data warehouse. Specialized data marts for different organizations departments, for instance, the finance department are then formed from there. The strength of this method is that it offers a clear structure for managing data, however, this method can be expensive as well as time-consuming and for that reason, it is ideal for large organizations only.



The essential components are discussed below:

1. **External Sources:** External source is a source from where data is collected irrespective of the type of data. Data can be structured, semi

structured and unstructured as well.

2. **Stage Area:** Since the data, extracted from the external sources does not follow a particular format, so there is a need to validate this data to load into datawarehouse. For this purpose, it is recommended to use **ETL** tool.

- **E(Extracted):** Data is extracted from External data source.
- **T(Transform):** Data is transformed into the standard format.
- **L(Load):** Data is loaded into datawarehouse after transforming it into the standard format.

3. **Data-warehouse:** After cleansing of data, it is stored in the data warehouse as central repository. It actually stores the meta data and the actual data gets stored in the data marts. **Note** that data warehouse stores the data in its purest form in this top-down approach.

4. **Data Marts:** Data mart is also a part of storage component. It stores the information of a particular function of an organisation which is handled by single authority. There can be as many number of data marts in an organisation depending upon the functions. We can also say that data mart contains subset of the data stored in data warehouse.

5. **Data Mining:** The practice of analysing the big data present in data warehouse is [data mining](#). It is used to find the hidden patterns that are

present in the database or in data warehouse with the help of algorithm of data mining.

This approach is defined by **Inmon** as – data warehouse as a central repository for the complete organisation and data marts are created from it after the complete data warehouse has been created.

Advantages of Top-Down Approach

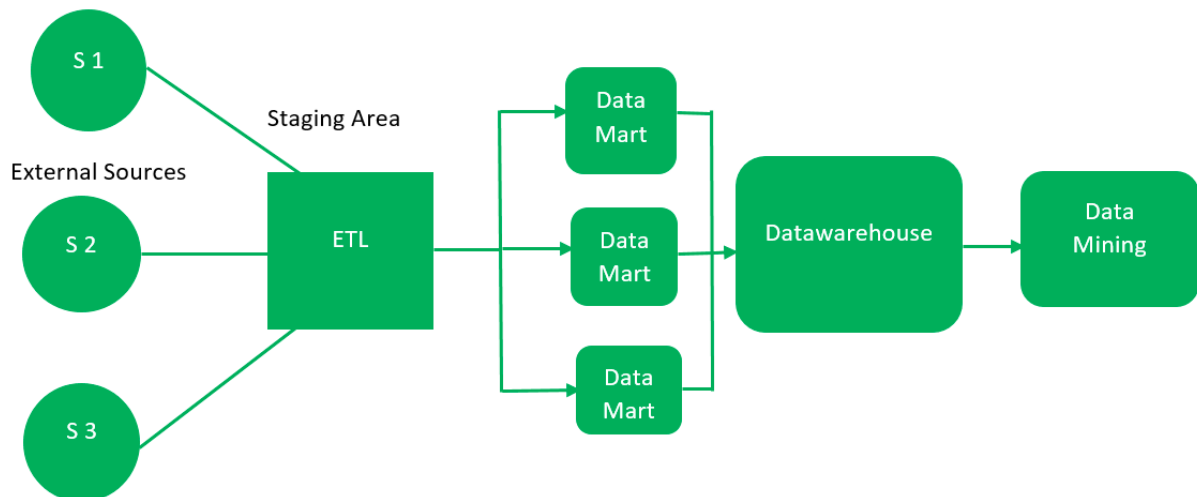
1. Since the data marts are created from the datawarehouse, provides consistent dimensional view of data marts.
2. Also, this model is considered as the strongest model for business changes. That's why, big organisations prefer to follow this approach.
3. Creating data mart from data warehouse is easy.
4. **Improved data consistency:** The top-down approach promotes data consistency by ensuring that all data marts are sourced from a common data warehouse. This ensures that all data is standardized, reducing the risk of errors and inconsistencies in reporting.
5. **Easier maintenance:** Since all data marts are sourced from a central data warehouse, it is easier to maintain and update the data in a top-down approach. Changes can be made to the data warehouse, and those changes will automatically propagate to all the data marts that rely on it.
6. **Better scalability:** The top-down approach is highly scalable, allowing organizations to add new data marts as needed without disrupting the existing infrastructure. This is particularly important for organizations that are experiencing rapid growth or have evolving business needs.
7. **Improved governance:** The top-down approach facilitates better governance by enabling centralized control of data access, security, and

quality. This ensures that all data is managed consistently and that it meets the organization's standards for quality and compliance.

8. **Reduced duplication:** The top-down approach reduces data duplication by ensuring that data is stored only once in the data warehouse. This saves storage space and reduces the risk of [data inconsistencies](#).
9. **Better reporting:** The top-down approach enables better reporting by providing a consistent view of data across all data marts. This makes it easier to create accurate and timely reports, which can improve decision-making and drive better business outcomes.
10. **Better data integration:** The top-down approach enables better data integration by ensuring that all data marts are sourced from a common data warehouse. This makes it easier to integrate data from different sources and provides a more complete view of the organization's data.

Bottom-Up Approach

Bottom up Approach is the Ralph Kimball's approach of the construction of individual data marts that lie at the center of specific business goals or functions such as marketing or sales. These data marts are extracted transformed & loaded first to provide organizations' ability to generate reports instantly. In turn, these data marts are affiliated to the more centralized and broad data warehouse system. This is a more flexible method of training, cheaper and best recommendable in smaller organizations.



1. First, the data is extracted from external sources (same as happens in top-down approach).
2. Then, the data go through the staging area (as explained above) and loaded into data marts instead of datawarehouse. The data marts are created first and provide reporting capability. It addresses a single business area.
3. These data marts are then integrated into datawarehouse.

This approach is given by **Kinball** as – data marts are created first and provides a thin view for analyses and datawarehouse is created after complete data marts have been created.

Advantages of Bottom-Up Approach

1. As the data marts are created first, so the reports are quickly generated.
2. We can accommodate more number of data marts here and in this way datawarehouse can be extended.
3. Also, the cost and time taken in designing this model is low comparatively.
4. **Incremental development:** The bottom-up approach supports incremental development, allowing for the creation of data marts one at a

time. This allows for quick wins and incremental improvements in data reporting and analysis.

5. **User involvement:** The bottom-up approach encourages user involvement in the design and implementation process. Business users can provide feedback on the data marts and reports, helping to ensure that the data marts meet their specific needs.
6. **Flexibility:** The bottom-up approach is more flexible than the top-down approach, as it allows for the creation of data marts based on specific business needs. This approach can be particularly useful for organizations that require a high degree of flexibility in their reporting and analysis.
7. **Faster time to value:** The bottom-up approach can deliver faster time to value, as the data marts can be created more quickly than a centralized data warehouse. This can be particularly useful for smaller organizations with limited resources.
8. **Reduced risk:** The bottom-up approach reduces the risk of failure, as data marts can be tested and refined before being incorporated into a larger data warehouse. This approach can also help to identify and address potential data quality issues early in the process.
9. **Scalability:** The bottom-up approach can be scaled up over time, as new data marts can be added as needed. This approach can be particularly useful for organizations that are growing rapidly or undergoing significant change.
10. **Data ownership:** The bottom-up approach can help to clarify data ownership and control, as each data mart is typically owned and managed by a specific business unit. This can help to ensure that data is accurate and up-to-date, and that it is being used in a consistent and appropriate way across the organization.

Metadata

Metadata is data about data which defines the data warehouse. It is used for building, maintaining and managing the data warehouse. In the Data Warehouse Architecture, meta-data plays an important role as it specifies the source, usage, values, and features of data warehouse data. It also defines how data can be changed and processed. It is closely connected to the data warehouse.

Metadata can be classified into following categories:

1. Technical Meta Data: This kind of Metadata contains information about warehouse which is used by Data warehouse designers and administrators.
2. Business Meta Data: This kind of Metadata contains detail that gives end-users a way easy to understand information stored in the data warehouse.

Data Marts

A data mart is an access layer which is used to get data out to the users. It is presented as an option for large size data warehouse as it takes less time and money to build. However, there is no standard definition of a data mart is differing from person to person. In a simple word **Data mart is a subsidiary** (subordinate) of a data warehouse. The data mart is used for partition of data which is created for the specific group of users. Data marts could be created in the same database as the Data warehouse or a physically separate Database.

OLAP

- Online Analytical Processing (OLAP) is a category of software that allows users to analyze information from multiple database systems at the same time.
- OLAP (Online Analytical Processing) is the technology behind many Business Intelligence (BI) applications.

- OLAP is a powerful technology for data discovery, including capabilities for limitless report viewing, complex analytical calculations, and predictive “what if” scenario (budget, forecast) planning.
- OLAP enables end-users to perform analysis of data in multiple dimensions, thereby providing the insight and understanding they need for better decision making.

Features of OLAP in data warehouse are the following:

- o Has intuitive easy-to-use interface;
- o OLAP supports complex calculations;
- o Provides data view in multidimensional manner;
- o Time intelligence.

The three major OLAP models in data warehouse:

- **ROLAP or Relational OLAP:**

The kind of system where users query data from a relational database or from their own local tables. Thus, the number of potential questions is not limited.

- **MOLAP or Multidimensional OLAP:**

- o This system stores the data in multidimensional database. Provides high speed of calculations.
- o This is the more traditional way of OLAP analysis.
- o The storage is not in the relational database, but in array-based structures formats.

- **HOLAP or Hybrid OLAP:**

- o It is a mix of above two mentioned systems. Pre-computed aggregates and cube structure stored in multidimensional database.

o HOLAP leverages cube technology for faster performance. It stores only the indexes and aggregations in the multidimensional form while the rest of the data is stored in the relational database.

Other Types

There are also less popular types of OLAP which are listed below.

Web-Enabled OLAP (WOLAP) Server

WOLAP pertains to OLAP application which is accessible via the web browser. Unlike traditional client/server OLAP applications, WOLAP is considered to have a three-tiered architecture which consists of three components: a client, a middleware, and a database server.

Desktop OLAP (DOLAP) Server

DOLAP permits a user to download a section of the data from the database or source, and work with that dataset locally, or on their desktop.

Mobile OLAP (MOLAP) Server

Mobile OLAP enables users to access and work on OLAP data and applications remotely through the use of their mobile devices.

Spatial OLAP (SOLAP) Server

SOLAP includes the capabilities of both Geographic Information Systems (GIS) and OLAP into a single user interface. It facilitates the management of both spatial and non-spatial data.

Need of OLAP

The key driver of OLAP is the multidimensional nature of the business problem. These problems are characterized by retrieving a very large number of

records that can reach gigabytes and terabytes and summarizing this data into form information that can be used by business analysts.

One of the limitations that SQL has, it cannot represent these complex problems.

A query will be translated into several SQL statements. These SQL statements will involve multiple joins, intermediate tables, sorting, aggregations and a huge temporary memory to store these tables. These procedures required a lot of computation which will require a long time in computing.

The second limitation of SQL is its inability to use mathematical models in these SQL statements. If an analyst, could create these complex statements using SQL statements, still there will be a large number of computation and huge memory needed. Therefore the use of OLAP is preferable to solve this kind of problem.

Characteristics of OLAP

The main characteristics of OLAP are as follows:

1. Multidimensional conceptual view: OLAP systems let business users have a dimensional and logical view of the data in the data warehouse. It helps in carrying slice and dice operations.
2. Multi-User Support: Since the OLAP techniques are shared, the OLAP operation should provide normal database operations, containing retrieval, update, adequacy control, integrity, and security.
3. Accessibility: OLAP acts as a mediator between data warehouses and front-end. The OLAP operations should be sitting between data sources (e.g., data warehouses) and an OLAP front-end.
4. Storing OLAP results: OLAP results are kept separate from data sources.

5. Uniform documenting performance: Increasing the number of dimensions or database size should not significantly degrade the reporting performance of the OLAP system.
6. OLAP provides for distinguishing between zero values and missing values so that aggregates are computed correctly.
7. OLAP system should ignore all missing values and compute correct aggregate values.
8. OLAP facilitate interactive query and complex analysis for the users.
9. OLAP allows users to drill down for greater details or roll up for aggregations of metrics along a single business dimension or across multiple dimensions.
10. OLAP provides the ability to perform intricate calculations and comparisons.
11. OLAP presents results in a number of meaningful ways, including charts and graphs.

Benefits of OLAP

OLAP holds several benefits for businesses: -

1. OLAP helps managers in decision-making through the multidimensional record views.
2. OLAP functions are self-sufficient owing to flexibility support to the organized databases.
3. It facilitates simulation of business models and problems, through extensive management of analysis-capabilities.
4. In conjunction with data warehouse, OLAP can be used to support a reduction in the application backlog, faster data retrieval, and reduction in query drag.

OLAP Operations

Since OLAP servers are based on multidimensional view of data, there are OLAP operations for multidimensional data.

The list of OLAP operations –

- o Roll-up
- o Drill-down
- o Slice and dice
- o Pivot (rotate)

Roll-up

Roll-up performs aggregation on a data cube in any of the following ways

- o By climbing up a concept hierarchy for a dimension
- o By dimension reduction
- o Roll-up is performed by climbing up a concept hierarchy for the dimension location.
- o Initially the concept hierarchy was “street < city < province < country”.
- o On rolling up, the data is aggregated by ascending the location hierarchy from the level of city to the level of country.
- o The data is grouped into cities rather than countries.
- o When roll-up is performed, one or more dimensions from the data cube are removed.

Drill-down

Drill-down is the reverse operation of roll-up. It is performed by either of the following ways –

- o By stepping down a concept hierarchy for a dimension
- o By introducing a new dimension.
- o Drill-down is performed by stepping down a concept hierarchy for the dimension time.
- o On drilling down, the time dimension is descended from the level of quarter to the level of month.
- o When drill-down is performed, one or more dimensions from the data cube are added.
- o It navigates the data from less detailed data to highly detailed data.

Slice

- The slice operation selects one particular dimension from a given cube and provides a new sub-cube.
- It will form a new sub-cube by selecting one or more dimensions.

Dice

- o Dice selects two or more dimensions from a given cube and provides a new sub-cube.
- o The dice operation on the cube based on the selection criteria.

Pivot

- o The pivot operation is also known as rotation. It rotates the data axes in view in order to provide an alternative presentation of data.

The advantages of OLAP are as follows –

- Business-centred multidimensional information.

- Business-centred figuring's.
- Dependable information and figuring's.
- Speed-of-thought examination.
- Adaptable, self-administration detailing.

Disadvantages

The disadvantages of OLAP are as follows –

- Pre-demonstrating is an absolute necessity
- Extraordinary reliance on IT.
- Helpless calculation capacity.
- Shy Interactive examination capacity.
- Slow in responding.
- Theoretical model.
- Extraordinary, expected danger.

Section A:

1. Define Data warehousing.
2. Name the characteristics of data warehousing.
3. Define ETL.
4. Name the steps to build the data warehousing.
5. List out the operations in OLAP.
6. What is data mart.
7. Define metadata.
8. Define OLAP.
9. What is MOLAP.
10. What is WOLAP.

11. Define SOLAP.
12. Name the features of OLAP.
13. What is Roll-up.
14. Define Drill-Down.
15. Define staging area.

SECTION B:

1. Explain what is data warehousing and mention the characteristics of data warehousing.
2. Briefly explain the advantages and disadvantages of OLAP.
3. Explain the list of operations in OLAP.
4. Briefly explain the top-down architecture of data warehousing.
5. Explain any three types of OLAP.
6. Write short notes on the characteristics of OLAP.
7. Explain the need and benefits of OLAP.
8. Write short notes on Metadata and Datamart.
9. Briefly explain the Advantages of Top-Down Approach.
10. Briefly explain the bottom-up architecture of data warehousing

Section C

1. Explain the types of architecture of data warehousing.
2. Explain how to build a data warehouse in detail.
3. Explain OLAP and its types in detail,
4. Explain the benefits and characteristics of OLAP.
5. Explain OLAP operations with its advantages and disadvantages of OLAP.